TECHNICAL & COST AUDIT · 29 OCTOBER 2026

Can Ask KiqStat pay for itself when every subscriber uses all of it?

At normal use, yes, on every plan, with room to spare. At the ceiling the code allows today, no: a single question can cost about 27 times a normal one. Separately, Ask KiqStat has been down in production since around 10 October because the Anthropic account ran out of credit.

MEASURED read from the database, production logs or the code

ESTIMATED worked out from prompt sizes and published prices; not yet checked against an invoice

DOWN	Every question since about 10 October fails with Anthropic's “credit balance is too low” error. Users see “Ask KiqStat is busy” and retry. MEASURED
₦2 HOLDS	A typical question costs about **₦2** (US$0.0012) on Claude Haiku 5.5. The “₦50–100” figure in the code comment is out of date. ESTIMATED
ENFORCED	Daily limits (free 3, Pro 30, VIP 100) are claimed in the database before the model runs, in one atomic step, so parallel requests can't slip past them. MEASURED
EXPOSED	The worst case is uncapped in practice: up to 6 model calls of up to 8,000 output tokens each, about **₦53** per question. At that rate a VIP member at full allowance costs ₦159,000 a month. ESTIMATED
BLIND	No token counts are stored anywhere, so per-question cost can't be measured from our side, and the credit running out went unnoticed for 2½ weeks. MEASURED

1. How it works

- **Provider and model:** Anthropic, `claude-haiku-5-5`, called through the official SDK and streamed to the browser. MEASURED src/app/api/ask/route.ts:33

- **Settings:** effort `medium` (thinking on, billed as output), `max_tokens: 8000` per call, up to **6 calls** per question (tool rounds, plus at most 2 re-issues for a malformed tool call within those 6). MEASURED route.ts:35, 177, 181, 206
- **Prompt:** a fixed system prompt (3,314 characters) and six tool definitions (about 3,950 characters of source). The date, time zone, open match and the user's combo go in the user's message, not the system prompt, so the fixed part never changes and can be cached. MEASURED src/lib/ask/prompt.ts, src/lib/ask/request.ts:171
- **Conversation:** the last 12 turns are re-sent each time, questions capped at 500 characters and earlier answers at 2,500. MEASURED request.ts:48–50
- **Data:** the model never sees raw data. Six tools (search fixtures, read a prediction, check prices, top picks, live scores, show picks) return compact JSON from the same cached predictions the site uses. Free users get a reduced view with a note about what is Pro. MEASURED src/lib/ask/tools.ts

2. What a question costs

Prices for Claude Haiku 5.5 (prompts up to 100K tokens): input US$0.10 per million tokens, output US$0.50, cache write US$0.125 (1.25×), cache read US$0.01 (0.1×). MEASURED from Anthropic's published price list. Naira figures use **₦1,600 per US$**; change it if your rate differs.

Token counts below come from the prompt and tool sizes, at about 4 characters per token plus 30% for Haiku 5.5's newer tokenizer. The fixed part (system prompt, tools, tool overhead) comes to about 2,400 tokens, above the 512-token minimum needed for caching. ESTIMATED

QUESTION	WHAT HAPPENS	INPUT (FRESH / CACHED)	OUTPUT INCL. THINKING	US$	₦
Typical	3 calls: find the match, read the prediction, answer in about 90 words	4,400 / 8,250	1,200	0.0012	2.0
Heavy	6 calls: combo of 8 top picks, prices checked, cards shown	10,000 / 40,000	3,000	0.0032	5.1
Ceiling	6 calls, each writing the full 8,000 tokens, nothing cached	90,000 / 0	48,000	0.033	53

All three rows are estimates. The typical row is the one to check against Anthropic's usage page for 9–10 October, the last days questions were answered (see section 6). The ceiling is unlikely, since answers are told to stay under 90 words, but nothing in the code prevents it.

3. Usage limits

- **Accounts:** `ask_claim` adds one to today's count only if it is still under the limit, in a single database statement, before the model is called. Two requests at once can't both take the last question. The day follows Lagos time. MEASURED supabase/migrations/0044_advisor_fixes.sql
- **Visitors without an account:** 2 questions per browser for good (stored as a hash of a cookie), plus 40 a day per network address as a backstop for cleared cookies. MEASURED src/lib/ask/guest.ts
- **Refunds:** when no answer arrives, the question is handed back. The refund functions can only be called by the server; an earlier version that browsers could call was closed in migration 0044. MEASURED
- **Passes get VIP's 100 a day.** A ₦500 day pass makes the account VIP, so it gets VIP's allowance: 100 questions for ₦500. MEASURED route.ts:104
- **The copy says "unlimited".** The free-limit message says "Pro makes it unlimited", and the panel hides the paid caps. Pro is 30 a day. MEASURED route.ts:116, 129

4. Cost controls in place, and missing

CONTROL	STATE	DETAIL
Cheapest capable model	In place	Haiku 5.5, about 1/20 of Sonnet 5.5 per token.
Prompt caching	In place	The system prompt and tools are cached, and so is the growing conversation inside one question. Nothing changing (like the date) sits in the cached part.
Compact tool data	In place	Tools return trimmed JSON from cached predictions (14 selections at most, 5 form games, 12 fixtures). The data lookups cost no tokens.
Output ceiling	**Too loose**	`max_tokens` is 8,000 per call while answers are meant to be under 90 words. This is the biggest gap between typical and worst-case cost.
Effort level	**Untested**	`medium` thinking on every call. Chat questions often answer just as well at `low`, with less thinking billed. Needs a quality check before changing.

Model routing	Not needed	Everything already runs on the cheapest model.
Reusing answers	None	Common questions (“top picks today”) are answered from scratch each time. Low priority at today's volume.
Usage logging and alerts	**Missing**	Token counts in each response are thrown away. No spend alert, no error-rate alert.

5. Failure handling and surprise charges

SCENARIO	WHAT HAPPENS	RISK
Parallel or double-submitted questions	Each takes a question from the allowance atomically. The panel blocks a second send while one is running; other tabs or scripts can send in parallel but still stay within the daily limit.	Low
Anthropic errors (overload, 5xx)	The SDK retries up to 2 times on its own. A failed request is generally not billed. The question is refunded if no answer came.	Low
Malformed tool call	Re-issued up to 2 times, counted inside the 6-call cap.	Low
User presses Stop before any text	The model call is cancelled, so we pay only for tokens already generated, but the question is **refunded**. Someone could ask, stop, and repeat without ever using up their allowance, paying nothing while we pay each time.	**Medium**
Failure after a few tool rounds, before any text	We have paid for those rounds and the question is refunded. Fair to the user; we absorb the cost.	Low
Credit runs out	Every call fails, every question is refunded, users see “busy” and retry. No alert reached anyone. This is happening now.	**High**

6. What we can measure

MEASURE	VALUE	SOURCE
Questions answered for signed-in users since 9 Oct	19	database, daily counts
Signed-in users who used it	40	database
Highest count for one user in one day	3	database (free limit)

Average per active user-day (free)	0.45	database
Pro / VIP questions answered	0	database
Questions answered for visitors without an account	74	database, 54 browsers
User-days where every question was refunded	37 of 46	database
Last day a signed-in question was answered	10 Oct	database
Cause of failures 26–28 Oct	credit balance too low	production logs, every POST /api/ask
Tokens per question	not recorded	needs Anthropic's usage page, or fix 2 below

Counts only; no user details were read or included. To get a measured cost per question: take the total spend on Anthropic's Usage page from 9 to 10 October and divide it by the questions answered then (19 signed-in, plus the visitors' share of 74).

7. Full monthly use per subscriber

Every subscriber asks their full daily allowance every day for 30 days. Revenue is the monthly price before Paystack's fee (1.5% + ₦100). ESTIMATED

PLAN	QUESTIONS / MONTH	PRICE	COST, TYPICAL	COST, HEAVY	COST, CEILING
Free	90	₦0	₦180	₦460	₦4,770
Pro (monthly)	900	₦5,000	₦1,800 · 36%	**₦4,600 · 92%**	**₦47,700**
VIP (monthly)	3,000	₦12,000	₦6,000 · 50%	**₦15,400 · 128%**	**₦159,000**
VIP day pass	100	₦500	₦200 · 39%	**₦510 · 102%**	**₦5,300**

Percentages are cost as a share of the price. Yearly and 3-month plans cost less per month (Pro yearly is ₦4,000 a month), so their margins are thinner than shown. Real use is far below these limits today (0.45 questions per active day), so this is the exposure, not the forecast.

8. Recommended fixes, in order

1. **Top up the Anthropic credit and turn on auto-reload and a monthly spend limit** (Anthropic Console → Billing). This is a settings change for the account owner, not code. At about ₦2 a question, US$20 covers roughly 16,000 questions.

2. **Log tokens for every question.** Store the input, cached, output tokens and cost for each answer in a small table, and show cost per question and per plan in admin. This turns every estimate here into a measurement.

3. **Alert on failures.** Email the team when more than a few questions in a row fail, and show users "Ask KiqStat is unavailable" instead of "busy" when the cause is billing, so they stop retrying.

4. **Lower `max_tokens` from 8,000 to about 1,500 per call, and the call cap from 6 to 4.** The ceiling falls from about ₦53 to about ₦14 a question, with no change to normal answers, which use a few hundred tokens.

5. **Give passes their own allowance.** A ₦500 pass should get about 30 questions, not VIP's 100.

6. **Don't refund a question the user stopped** once the model has started. Keep refunds for failures on our side.

7. **Fix the "unlimited" wording.** Say "Pro: 30 questions a day", so nobody is promised more than they get.

8. **Test effort `low` against `medium`** on 30–50 real questions once logging is in. If the answers hold up, thinking costs fall further.

9. **Revisit the VIP cap once there is real data.** VIP (₦400 a day) breaks even at about 78 heavy questions a day, or about 28 a day at the new ceiling from fix 4. If measured use stays near today's, 100 a day is safe; if heavy users appear, 60 a day keeps every VIP member profitable at heavy use.

Files reviewed

- `src/app/api/ask/route.ts`
- `src/lib/ask/prompt.ts`
- `src/lib/ask/request.ts`
- `src/lib/ask/tools.ts`

- src/lib/ask/guest.ts
- src/components/ask/ask-panel.tsx
- supabase/migrations/0026, 0027, 0044

No API keys, credentials or customer details are in this report.